

JUAN MIGUEL CRUZ

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🌐 https://jcruz202.github.io/Juan_Cruz_Personal_Portfolio_Website/index.html

Education

University of Illinois Chicago (UIC)

Expected graduation: May 2025

Bachelor of Science in Computer Science – Software Engineering Concentration

• **Extracurricular:** Robotics Club - Fall 2021, Campus Outreach - Fall 2021

Technical Skills

Programming Language: Python, C++, C, Java, HTML/CSS, SQL, JavaScript, Fortran, Ada

Certifications: Responsive Web Design - June 2023, AWS Cloud Practitioner - October 2024

Security Clearance: Secret, Interim Top secret

Technical Experience

Northrop Grumman - Associate Software Engineer

December 2025 - Present

- Rapidly acquired proficiency in Fortran, Ada, and a legacy codebase of more than 1.2 million lines within two months, then contributed to development of multiple subsystems within the larger software system.
- Built working knowledge of Air Force mission operations to support development of simulation software that models real-world scenarios, translating operational needs into fixes and enhancements for more reliable ticket resolution.
- Earned a BRAVO Award within 6 months for proactively resolving a critical ticket three days before the code deadline, helping keep the release on schedule.

Headstarter - Software Engineering Fellowship

July 2024 - September 2024

- Collaborated in a team of 3 to build and deploy 5 AI projects in 5 weeks using React JS, Next.js, Firebase, Clerk, and Vercel, following agile methodologies with weekly sprints and incorporating CI/CD practices for iterative deployment.
 - Implemented a web scraper to extract data from Rate My Professor and upsert it to a Pinecone index, integrated with a RAG pipeline using LangChain and OpenAI GPT-4o for up-to-date answers to user queries.
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Projects

AI Flashcards SaaS - Fellowship | [Github Repo](#)

August 2024

- Developed a SaaS Flashcards app using JavaScript, integrating 3 key technologies: OpenAI for adaptive content generation, Clerk for authentication, and Firebase for data storage as part of a 7 week fellowship program.

LinkUp - Personal | [Github Repo](#)

September 2024

- Constructed the architecture for a full-stack mobile application as a personal project using JavaScript and AWS Amplify, integrating AWS Cognito for authentication, S3 for storage, and DynamoDB for database management.
- Enhanced user experience with an AI-powered customer support system using OpenAI, demonstrating integration of advanced technologies in practical applications.

Message App | [Github Repo](#)

April 2023

- Crafted a Java-based messaging application emulating iPhone functionality, leveraging server sockets to support concurrent multi-client connections.
 - Utilized threading and networking protocols to enable targeted individual messaging and selective group communication features for 3+ concurrent users.
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Leadership Involvement

Society of Hispanic Professional Engineers - Historian & Mentor *SHPE Mentor*

January 2024 - May 2025

- Optimized alumni database management, improving data accessibility and tracking for 100+ members
- Restructured Drive storage, reducing file search time, and enhanced navigation efficiency for board members.

Ignite UIC Leadership Program - Mentor

September 2022

- Attained a 6-week leadership training, mentoring 3 freshmen on core leadership principles with 100% program completion rate.
- Facilitated weekly sessions and supported freshmen in building connections and enhancing their leadership abilities.